

Monitoring Task

Task Description:

Install Prometheus and Grafana on a Linux EC2 machine, connect Prometheus to Grafana, and create a dashboard to view metrics.

Provisioned EC2 instance with terraform

```

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS C:\Users\Kousick Devaraj\Desktop\DevOps\terraform\terraform-ec2-instance> cat .\main.tf
provider "aws" {
    region = "ap-south-1"
}

resource "aws_instance" "monitoring-server" {
    ami= "ami-01a00762f46d584a1"
    instance_type = "t2.medium"

    tags = {
        Name = "Observability-server"
    }
}
PS C:\Users\Kousick Devaraj\Desktop\DevOps\terraform\terraform-ec2-instance>

```

```

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.monitoring-server: Creating...
aws_instance.monitoring-server: Still creating... [00m10s elapsed]
aws_instance.monitoring-server: Creation complete after 13s [id=i-0e7c523011eb37d99]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS C:\Users\Kousick Devaraj\Desktop\DevOps\terraform\terraform-ec2-instance> |

```

Instances (1/1) Info Last updated about 1 hour ago

Saved filter sets
Choose filter set

☒ Find Instance by attribute or tag (case-sensitive)

Instance state = running X Clear filters ▼

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...
☒	Observability-server	i-0e7c523011eb37d99	Running	t2.medium	2/2 checks passed	ap-south-1b	ec2-3-109-213-186.ap-...	3.109.213.186

Inbound rules (7) Manage tags Edit inbound rules

Search

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-01aae0bf6f681fa36	IPv4	HTTP	TCP	80	0.0.0.0/0
☐	-	sgr-00c528c55f21ddfa0	IPv4	Custom TCP	TCP	9093	0.0.0.0/0
☐	-	sgr-0b06a2cc572417808	IPv4	Custom TCP	TCP	3000	0.0.0.0/0
☐	-	sgr-02973c753194957a9	IPv4	Custom TCP	TCP	9100	0.0.0.0/0
☐	-	sgr-0817a054ad280e97a	IPv4	SSH	TCP	22	0.0.0.0/0
☐	-	sgr-0eff7cc8211d117e7	IPv4	HTTPS	TCP	443	0.0.0.0/0
☐	-	sqr-0471a27b56e515d7c	IPv4	Custom TCP	TCP	9090	0.0.0.0/0

Installing prometheus and grafana using docker compose:

```
ubuntu@ip-172-31-2-73:~/monitoring$ ls -ll
total 8
-rwxr-xr-x 1 ubuntu ubuntu 222 Aug 21 07:39 prometheus.yml
-rwxr-xr-x 1 ubuntu ubuntu 689 Aug 21 07:48 docker-compose.yml
ubuntu@ip-172-31-2-73:~/monitoring$
ubuntu@ip-172-31-2-73:~/monitoring$
```

```
ubuntu@ip-172-31-2-73:~/monitoring$ docker compose up -d
[+] Running 26/26
✓ grafana Pulled
  ✓ 23a073b63fe4 Pull complete
  ✓ 55afalecc21d Pull complete
  ✓ 6e5f17902521 Pull complete
  ✓ 530039403505 Pull complete
  ✓ adfbc35548cc Pull complete
  ✓ d563fee7e84d Pull complete
  ✓ 9baf42ea803d Pull complete
  ✓ 7e96b853aec8 Pull complete
  ✓ feefe83fec6c Pull complete
  ✓ 3395a07b0f7d Pull complete
  ✓ 4f4fb700ef54 Pull complete
✓ node_exporter Pulled
  ✓ f5ee56b8245c Pull complete
  ✓ c93f261857c3 Pull complete
  ✓ bef5f3176279 Pull complete
✓ prometheus Pulled
  ✓ 487858d29734 Pull complete
  ✓ da388505dad8 Pull complete
  ✓ e80cad428b4d Pull complete
  ✓ 200d6e325877 Pull complete
  ✓ 49c226752edd Pull complete
  ✓ 8ee344dffb17 Pull complete
  ✓ 7a5b1d5c3c45 Pull complete
  ✓ eb7a272b9937 Pull complete
  ✓ 0dc8fee0e5ee Pull complete
[+] Running 6/6
✓ Network monitoring_default Created
✓ Volume monitoring_grafana_data Created
✓ Volume monitoring_prometheus_data Created
✓ Container prometheus Started
✓ Container grafana Started
✓ Container node_exporter Started
ubuntu@ip-172-31-2-73:~/monitoring$
```

```
ubuntu@ip-172-31-2-73:~/monitoring$ docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
grafana	grafana/grafana:latest	"/run.sh"	grafana	6 minutes ago	Up 6 minutes	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
node_exporter	prom/node-exporter:latest	"/bin/node_exporter"	node_exporter	6 minutes ago	Up 6 minutes	0.0.0.0:9100->9100/tcp, [::]:9100->9100/tcp
prometheus	prom/prometheus:latest	"/bin/prometheus --c..."	prometheus	6 minutes ago	Up 6 minutes	0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp

```
ubuntu@ip-172-31-2-73:~/monitoring$ more prometheus.yml
```

Browser screenshot of Prometheus Targets page (3.109.213.186:9090/targets). The page shows two targets:

Target	Endpoint	Labels	Last scrape	State
node_exporter	http://node_exporter:9100/metrics	instance="node_exporter:9100" job="node_exporter"	7.803s ago 9ms	UP
prometheus	http://localhost:9090/metrics	instance="localhost:9090" job="prometheus"	3.928s ago 4ms	UP

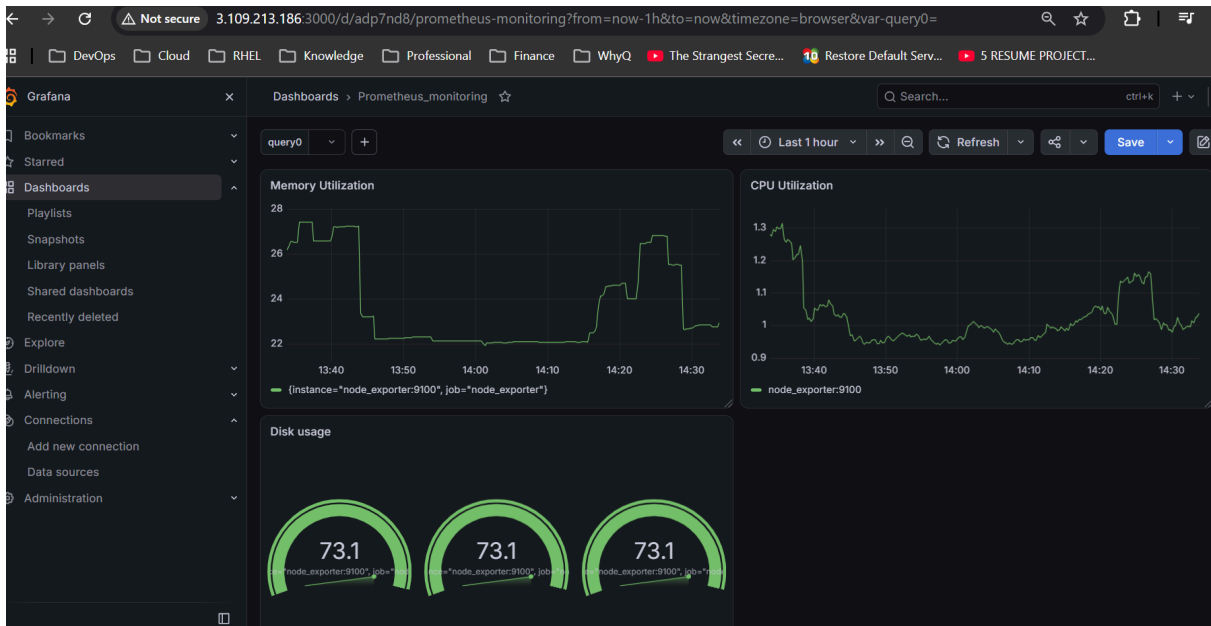
Browser screenshot of Prometheus Query page (3.109.213.186:9090/query). The query `up` is entered in the query box. The results are displayed in a table:

Series	Value
up{instance="node_exporter:9100", job="node_exporter"}	1
up{instance="localhost:9090", job="prometheus"}	1

Browser screenshot of Prometheus Query page (3.109.213.186:9090/query). The query `node_memory_MemTotal_bytes` is entered in the query box. The results are displayed in a table:

Series	Value
node_memory_MemTotal_bytes{instance="node_exporter:9100", job="node_exporter"}	4096536576

Grafana dashboard creation for monitoring prometheus server:



Not secure 3.109.213.186:3000/login

DevOps Cloud RHEL Knowledge Professional Finance WhyQ The Strangest Secre... Restore Default Serv... 5 RESUME PROJECT...

Welcome to Grafana

Email or username

email or username

Password

password

Log in

Forgot your password?

← → ↻ ⚠ Not secure 3.109.213.186:3000/connections/datasources/edit/cfvtdi7omsxsc

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Grafana ×

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➕ Add new connection

Data sources

⚙ Administration ▾

Connections > Data sources > prometheus

🔍 Search... ctrl+k + ▾ ⓘ 🛠

Series limit ⓘ 40000

Use series endpoint ⓘ ☐

Exemplars

+ Add

✓ Successfully queried the Prometheus API.

Next, you can start to visualize data by [building a dashboard from scratch](#) or by querying data in the [Explore view](#).
[Open in Metrics Drilldown](#)